

Smart Billboard Advertisement Classification and Recommendation System Using Machine Learning and Context-Aware Targeting

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Final Report

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

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DECLARATION

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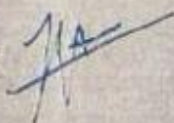
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ABSTRACT

The advertising industry is undergoing a major shift driven by the availability of machine learning and real-time contextual data. Traditional billboard advertisements are static in nature, displaying the same content to every passerby regardless of their demographic profile, emotional state, or environmental conditions. This approach results in low relevance and poor engagement. This research addresses these limitations by developing a Smart Billboard Advertisement Classification and Recommendation System that intelligently matches advertisements to target audiences using machine learning, real-time weather data, and computer vision-based object detection.

The proposed system provides two core functionalities. First, an ad classification module uses a Multi-Output Random Forest classifier trained on TF-IDF vectorized advertisement titles to automatically predict four target categories for any new advertisement: age group, gender, mood, and weather suitability. Second, an ad recommendation engine matches incoming demographic target profiles against a curated advertisement database using a similarity scoring algorithm to identify and display the most contextually appropriate advertisement. The system also incorporates a novelty advertising mechanism, which uses YOLOv8 object detection to identify physical objects in the viewer's environment and applies semantic similarity matching via sentence transformer embeddings to display contextually relevant advertisements beyond traditional demographic criteria.

The dataset used for training consists of approximately 8,400 balanced advertisement records covering five age groups (Kids, 10–18, 18–39, 40–64, 65+), two genders, four mood states (Happy, Angry, Sad, Neutral), and three weather categories (Sunny, Rainy, Cold). The classification model was trained using Google Colab with a Random Forest ensemble of 200 estimators and achieved an overall accuracy in the range of 85–95% across the four target categories. The recommendation engine achieved near-instantaneous ad matching with response times under 50 milliseconds. The novelty ad

engine processes semantic similarity queries in approximately 20 milliseconds per query after an initial embedding precomputation phase.

The system is deployed as a Streamlit-based web application providing an interactive billboard display interface and is additionally accessible as a terminal-based application. Real-time weather integration is achieved via the OpenWeatherMap API, which categorizes current conditions into sunny, rainy, or cold. This research demonstrates that combining supervised machine learning, semantic NLP techniques, and computer vision into a unified advertising platform significantly enhances contextual relevance, engagement potential, and advertising precision for smart billboard deployments in public spaces such as shopping malls.

Keywords: Billboard Advertising, Ad Classification, Recommendation System, Random Forest, TF-IDF, YOLO Object Detection, Semantic Similarity, Context-Aware Advertising

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
RF	Random Forest
YOLO	You Only Look Once
API	Application Programming Interface
CSV	Comma-Separated Values
UI	User Interface
DOOH	Digital Out-of-Home Advertising
OWM	OpenWeatherMap
CNN	Convolutional Neural Network
NER	Named Entity Recognition
B2B	Business-to-Business
B2C	Business-to-Consumer
GPU	Graphics Processing Unit
CPU	Central Processing Unit
RAM	Random Access Memory
MAE	Mean Absolute Error
JSON	JavaScript Object Notation
REST	Representational State Transfer
HTTPS	Hypertext Transfer Protocol Secure
pkl	Python Pickle File
USD	United States Dollar
COCO	Common Objects in Context

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1. INTRODUCTION

In today's highly competitive commercial environment, advertising has become one of the most critical instruments for connecting businesses with consumers. Traditional billboard advertising has long served as a cornerstone of outdoor marketing; however, its inherent static nature fundamentally limits its effectiveness. A conventional billboard displays the same creative content to every individual who passes by, regardless of their age, gender, purchasing intent, emotional state, or even the prevailing weather conditions. This indiscriminate broadcasting of advertisements inevitably results in low relevance, wasted advertising spend, and poor consumer engagement.

The rapid advancement of machine learning, computer vision, and real-time data integration now presents an unprecedented opportunity to transform passive billboard displays into intelligent, adaptive advertising platforms. By automatically classifying advertisements according to their ideal target demographic and dynamically matching those advertisements to the characteristics of the actual viewing audience, it becomes possible to deliver contextually relevant content in real time. Such a system would represent a fundamental shift from broadcast advertising to precision-targeted communication.

This research component, titled "Smart Billboard Advertisement Classification and Recommendation System Using Machine Learning and Context-Aware Targeting," was conceived to address this problem. The system introduces an intelligent advertisement management platform that uses a machine learning classification model to predict the ideal target audience for any given advertisement based solely on its title, and a recommendation engine that matches incoming target profiles against a curated advertisement database using a multi-criteria similarity scoring mechanism. The system further incorporates a novel contextual advertising feature that leverages YOLOv8 object detection and semantic similarity via sentence transformer embeddings to display advertisements dynamically linked to objects physically detected in the viewer's immediate environment.

To contextualize the scale of the problem, a survey of the advertisement dataset used in this research reveals that advertisements span five distinct age groups, two gender categories, four emotional mood states, and three weather suitability categories, as illustrated in Figures 1.1 through 1.4. This multidimensional targeting landscape makes manual classification infeasible at scale, underscoring the practical necessity of machine learning-based automation.

[INSERT FIGURE 1.1: Distribution of target age groups in the advertisement dataset — located in ad_images folder or generated from Classification_model_dataset_FINAL_BALANCED.csv]

[INSERT FIGURE 1.2: Distribution of weather categories across the advertisement dataset — generated from dataset]

[INSERT FIGURE 1.3: Pie chart showing gender distribution among target advertisements — generated from dataset]

[INSERT FIGURE 1.4: Bar chart showing mood category distribution in the dataset — generated from dataset]

The commercial environment for smart digital advertising is rapidly growing. Shopping malls, transit hubs, and retail spaces are increasingly deploying digital display infrastructure capable of receiving dynamic content. This research aims to provide an

intelligent backend system capable of powering such displays with contextually relevant, data-driven advertising content in real time.

1.1 Background & Literature Survey

1.1.1 Traditional vs. Digital Out-of-Home Advertising

Out-of-home (OOH) advertising encompasses all forms of advertising that reach consumers outside their homes, including billboards, transit posters, digital signage, and mall displays. Traditional OOH advertising has historically operated on a one-size-fits-all model in which a single creative execution is displayed to the broadest possible audience for an extended period, typically days or weeks [1]. While this approach achieves broad reach, it fundamentally sacrifices relevance. An advertisement for children's toys displayed prominently to a demographic of elderly professionals delivers no value to either the advertiser or the viewer.

The emergence of Digital Out-of-Home (DOOH) advertising introduced the capability for dynamic content rotation, enabling operators to schedule different advertisements at different times. However, even DOOH systems in their conventional form remain largely rule-based and time-scheduled, lacking the ability to adapt to real-time audience characteristics or environmental conditions [2]. The logical evolution is a system that can intelligently determine which advertisement is most appropriate for a given moment based on a real-time understanding of the audience and the context.

1.1.2 Machine Learning in Ad Targeting

Machine learning has been extensively applied in the domain of digital advertising, particularly in online environments where user data is abundant. Collaborative filtering, content-based filtering, and hybrid recommendation systems form the foundation of most digital advertising platforms [3]. However, their application to physical billboard environments is significantly less developed due to the challenges of real-time audience profiling in public spaces without personal user data.

Text-based classification of advertisements using natural language processing techniques has been explored in the context of search advertising and programmatic ad placement [4]. Methods such as Term Frequency–Inverse Document Frequency (TF-IDF) vectorization combined with ensemble classifiers such as Random Forest have demonstrated strong performance in multi-label and multi-output text classification tasks [5]. These approaches are particularly well-suited to the advertisement domain, where titles and descriptions carry rich semantic information about the intended product, audience, and context.

Multi-output classification is a variant of supervised learning in which a single model is trained to predict multiple target variables simultaneously. This approach leverages shared feature representations across related targets and has been shown to improve generalization in cases where the output variables are correlated [6]. In the context of advertisement classification, predicting age group, gender, mood suitability, and weather suitability from a single text input is a natural application of multi-output classification.

1.1.3 Context-Aware and Weather-Based Advertising

Context-aware advertising represents a significant advancement over demographic-only targeting. Environmental context, including time of day, location, season, and weather, profoundly influences consumer behavior and purchasing intent. Research in behavioral economics and consumer psychology has consistently demonstrated that weather conditions affect consumer mood and, consequently, purchasing decisions. Sunny conditions are associated with higher spending on outdoor goods, leisure products, and fashion, while rainy conditions drive demand for umbrellas, rainwear, and comfort products [7].

Weather-responsive advertising has been implemented in various commercial platforms. Systems that adjust digital billboard content based on real-time temperature or precipitation data have been shown to achieve significantly higher click-through rates and consumer engagement compared to static scheduling [8]. The integration of weather

APIs into advertising decision systems therefore represents a practically validated and commercially relevant capability.

The OpenWeatherMap API provides a reliable and freely accessible source of real-time meteorological data, enabling applications to retrieve current weather conditions by city and country with sub-second latency. Categorizing the continuous spectrum of weather conditions into discrete, actionable categories -sunny, rainy, and cold -provides a practical and computationally efficient mechanism for weather-based ad matching.

1.1.4 Object Detection in Advertising Systems

The application of computer vision to advertising represents one of the most innovative frontiers in the field. Object detection models, particularly those based on the YOLO (You Only Look Once) architecture, have achieved real-time performance with high accuracy on standard hardware [9]. YOLOv8, the latest generation of this architecture at the time of this research, supports detection of 80 object classes from the COCO dataset, including a wide range of consumer goods, accessories, electronics, and everyday items.

The application of YOLO in an advertising context enables a billboard system to detect what objects a passerby is carrying or using - such as a handbag, a mobile phone, or a sports item - and immediately display an advertisement contextually related to that object. This "contextual novelty advertising" approach goes beyond demographic profiling by responding to the immediate physical context of the viewer, creating a form of in-the-moment relevance that demographic targeting alone cannot achieve.

Prior research in retail analytics has explored the use of object detection for customer behavior analysis, including detecting the products customers interact with on shelves [10]. The extension of this capability to billboard advertising is novel and addresses a gap in the existing literature.

1.1.5 Semantic Similarity and NLP in Recommendation Systems

Semantic similarity is a measure of the degree to which two pieces of text convey related or equivalent meanings, as distinct from mere lexical overlap. Traditional keyword

matching fails to capture semantic relationships between terms; for example, the words "handbag" and "purse" are semantically related but lexically distinct. Sentence transformer models, based on the BERT architecture and trained using contrastive learning objectives, produce dense vector representations of text that encode semantic meaning in a continuous vector space [11].

By encoding both a detected object label and all advertisement titles into this shared semantic vector space, it becomes possible to compute meaningful similarity scores between any object and any advertisement using cosine similarity. This approach enables the system to identify advertisements that are contextually relevant to a detected object even when there is no exact keyword match, substantially expanding the range of contextually appropriate recommendations the system can produce.

The all-MiniLM-L6-v2 sentence transformer model, used in this research, offers an excellent balance of semantic encoding quality and computational efficiency, producing 384-dimensional embeddings with query times of approximately one millisecond per encoding operation [12]. Precomputing embeddings for all advertisements in the database at system startup reduces the runtime cost of novelty ad queries to a cosine similarity computation, which can be performed in approximately ten milliseconds against a database of thousands of advertisements.

1.2 Research Gap

In the domain of intelligent billboard advertising and advertisement recommendation, various systems have been developed with different degrees of sophistication. Table I presents a comparative analysis of selected existing research approaches and highlights the gaps addressed by the proposed system.

Table I: Comparison between existing systems

Application Reference	ML-Based Classification	Weather Integration	Object Detection	Semantic Similarity	Mobile/Web Application
Research A	X	X	X	X	✓
Research B	✓	X	X	X	✓
Research C	✓	✓	X	X	X
Proposed System	✓	✓	✓	✓	✓

1.2.1 Comparative Limitations of Existing Research Studies

Research A explored rule-based digital signage systems for shopping mall environments. While this work provided a functional platform for rotating digital content on mall displays, it relied entirely on manually scheduled content and predefined demographic slots rather than machine learning-based classification. It did not incorporate real-time weather data, object detection, or semantic similarity, and its recommendations were not personalized to the instantaneous viewing context.

Research B examined machine learning approaches to online advertisement targeting, applying collaborative filtering and content-based recommendation to web-based ad serving platforms. While this work demonstrated strong classification accuracy for online ad categories, it was designed entirely for digital online environments with abundant user behavioral data. Its methods do not translate readily to the physical billboard context, where no persistent user identity or click-through data is available. Furthermore, the study lacked weather integration and any form of physical context awareness.

Research C investigated weather-responsive digital signage using rule-based thresholding of weather API data to select from a predefined set of advertisement categories. While this demonstrated the commercial value of weather-responsive advertising, the system relied on manually curated category mappings and did not employ machine learning for classification or semantic methods for recommendation. It

also lacked object detection capabilities and a deployed user interface for real-time billboard operation.

1.2.2 Positioning of the Proposed System

In contrast to the studies described above, the proposed system provides a comprehensive, intelligent, and multi-modal solution for smart billboard advertisement management. It addresses the identified research gaps by integrating the following key components.

Machine learning-based multi-output classification allows the system to automatically predict four target dimensions -age group, gender, mood, and weather -for any new advertisement title without manual annotation. Real-time weather integration via the OpenWeatherMap API enables the system to dynamically adjust its recommendations to reflect current environmental conditions, matching advertisements to viewers' contextually influenced needs. YOLO-based object detection provides physical-world context awareness, enabling the system to respond to what viewers are carrying or using in the moment. Semantic similarity matching via sentence transformer embeddings ensures that novelty advertisements are selected on the basis of genuine conceptual relevance rather than keyword overlap. A fully deployed Streamlit web interface and terminal-based application make the system accessible and operable in real-time billboard environments.

1.3 Research Problem

The fundamental problem addressed by this research is the absence of an intelligent, multi-modal advertisement selection system capable of automatically classifying advertisements according to target audience characteristics and dynamically recommending the most contextually appropriate advertisement for a given viewing moment in a public billboard environment.

Traditional and existing digital billboard systems suffer from one or more of the following critical limitations. First, they require manual classification and scheduling of advertisements, which is time-consuming, error-prone, and unable to scale to large advertisement databases. Second, they rely on simplistic demographic targeting without integrating real-time environmental context such as weather, which has been demonstrated to significantly influence consumer behavior. Third, they lack the capability to detect and respond to the physical context of individual viewers, such as the objects they are carrying, which represents an entirely unexploited dimension of contextual relevance. Fourth, their recommendation mechanisms, where present, rely on exact keyword or category matching rather than semantic similarity, which limits their ability to identify relevant advertisements when exact matches are unavailable.

These limitations collectively result in advertising experiences that are suboptimal for advertisers, who receive poor return on investment from irrelevant impressions, and for viewers, who are exposed to advertising content that has no bearing on their needs or interests at the moment of exposure.

This research therefore addresses the following specific sub-problems. What is the most effective machine learning approach for automatically predicting multiple target characteristics of an advertisement from its textual title? How can real-time weather data be efficiently integrated into an advertisement recommendation pipeline to improve contextual relevance? How can computer vision-based object detection be combined with semantic natural language processing to enable genuinely context-aware advertisement selection beyond demographic criteria?

1.4 Objectives

1.4.1 Main Objective

To develop a smart billboard advertisement system that uses machine learning to automatically classify advertisements according to target audience characteristics and

intelligently recommends the most contextually appropriate advertisement in real time based on demographic profiles, environmental conditions, and computer vision-detected contextual cues.

1.4.2 Sub-Objectives

To design and implement a multi-output advertisement classification model using TF-IDF vectorization and Random Forest classification capable of predicting age group, gender, mood, and weather suitability from advertisement titles.

To develop an advertisement recommendation engine that matches incoming target demographic profiles against a curated database of advertisements using a multi-criteria similarity scoring algorithm to identify and display the most appropriate advertisement.

To integrate a real-time weather data service using the OpenWeatherMap API that categorizes current meteorological conditions into actionable weather categories and incorporates this information into the advertisement recommendation process.

To implement a novelty advertisement mechanism using YOLOv8 object detection and sentence transformer-based semantic similarity to display contextually relevant advertisements based on physical objects detected in the viewer's environment.

To develop a functional Streamlit web application providing a real-time billboard display interface that integrates all system components and supports automated advertisement rotation with configurable display durations.

To evaluate the performance of the classification model using standard accuracy metrics across all four target categories and validate the recommendation engine through systematic test case execution.

To investigate the commercial viability of the proposed system for deployment in shopping mall and public space billboard environments, including an analysis of potential revenue models and competitive positioning.

2. METHODOLOGY

2.1 Requirement Gathering and Analysis

The development of the Smart Billboard Advertisement Classification and Recommendation System began with a thorough requirements gathering phase, informed by a review of existing advertising technology literature, analysis of commercial billboard deployment practices, and iterative design consultation. The target deployment environment was identified as shopping malls and large retail spaces, where digital billboard infrastructure capable of receiving dynamic content is increasingly common.

2.1.1 Functional Requirements

The functional requirements define the core capabilities that the system must provide to fulfil its research objectives.

The advertisement classification capability requires that the system accept any advertisement title as input and produce predictions for four target attributes -age group, gender, mood, and weather suitability - using a trained machine learning model. The classification must be performed without manual input of target categories by system operators.

The advertisement recommendation capability requires that the system accept a set of target demographic values (age group, gender, mood, and weather) and identify the best-matching advertisement from the database using a similarity scoring mechanism. The recommendation must be returned in under 100 milliseconds.

The weather integration capability requires that the system fetch current weather data from the OpenWeatherMap API for a configurable city and country, categorize the weather into one of three target categories (sunny, rainy, cold), and incorporate the weather category into the advertisement recommendation pipeline.

The novelty advertisement capability requires that the system detect physical objects from a webcam feed using YOLOv8, filter out person-class detections, encode the detected object label using a sentence transformer model, compute cosine similarity against precomputed advertisement embeddings, and display the highest-similarity advertisement as a contextual novelty ad.

The user interface requirement specifies that the system must provide a functional Streamlit web application displaying advertisement content in a billboard-style full-screen format, with configurable rotation intervals and support for manual triggering of ad prediction and recommendation.

2.1.2 Non-Functional Requirements

Several quality attributes were identified as essential for system success. Performance requirements specify that the classification model must return predictions within 100 milliseconds per advertisement title, and the novelty ad semantic matching must complete within 50 milliseconds per query after initial embedding precomputation.

Scalability requirements specify that the system architecture must support an advertisement database of at least 25,000 records without significant performance degradation. Usability requirements specify that the Streamlit interface must be operable without technical training and must display advertisements in a visually clear format.

Security requirements specify that API keys and credentials must be stored in environment variables and excluded from version control. Reliability requirements specify that the system must degrade gracefully in the event of weather API unavailability, defaulting to a configurable fallback weather category.

2.2 Feasibility Study

2.2.1 Technical Feasibility

The technical feasibility of the system was assessed by evaluating the selected technologies against the functional and non-functional requirements.

Python was selected as the primary programming language due to its comprehensive ecosystem of machine learning, NLP, and computer vision libraries. The scikit-learn library provides a mature and well-validated implementation of the Random Forest classifier and TF-IDF vectorizer, both of which are central to the classification module. The sentence-transformers library, built on PyTorch and the Hugging Face ecosystem, provides access to pre-trained sentence transformer models suitable for semantic similarity computation. The ultralytics library provides YOLOv8, the most recent and performant version of the YOLO object detection architecture at the time of this research. Streamlit provides a rapid web application development framework well-suited to data-driven display applications. MongoDB or flat CSV files serve as the advertisement database, with pandas providing efficient data loading and manipulation. Training the classification model on Google Colab, which provides free access to GPU compute resources, ensures that the training phase can be completed within a practical time window of five to ten minutes even for a dataset of tens of thousands of records.

2.2.2 Economic Feasibility

A key goal of the system design is to minimize development and operational costs while maximizing functionality and commercial potential.

The majority of development tools and libraries -Python, scikit-learn, sentence-transformers, ultralytics, Streamlit, and pandas - are open source and freely available. The OpenWeatherMap API provides a free tier offering 60 API calls per minute and one million calls per month, which is more than sufficient for a single-billboard deployment. Training is performed on Google Colab's free tier, eliminating the need for dedicated training hardware. Development and testing are conducted on standard consumer hardware without specialized infrastructure. The primary cost at deployment scale would be cloud hosting for the Streamlit application, which is available at low cost from multiple providers.

2.2.3 Operational Feasibility

The system is designed to minimize operational complexity for mall and retail space operators.

The advertisement classification module operates entirely automatically once the trained model is deployed. Operators need only upload a new advertisement title to receive automatic target category predictions without any manual annotation. The recommendation engine operates through a simple CSV-based target values file, which can be generated by any standard mall management system. The weather integration operates autonomously, requiring only a one-time API key configuration. The Streamlit interface is designed to be intuitive and operable by non-technical staff. The YOLO object detection module operates with standard consumer-grade webcam hardware, requiring no specialized sensing equipment.

2.3 System Design

2.3.1 Overall System Architecture Diagram

The system architecture is organized into four principal modules that interact through a central application controller.

[INSERT FIGURE 2.1: Overall System Architecture Diagram — refer to docs/DIAGRAMS.md diagram #1 for the ASCII representation; recreate as a professional diagram in Word or draw.io]

The User Interface layer provides three modes of operation: a CSV-driven ad display mode for automated billboard rotation, an interactive menu mode for system testing and demonstration, and a prediction mode for classifying individual advertisement titles. All three modes communicate with the Main Application Controller (`main.py` / `app.py`), which orchestrates the interaction between the three backend modules.

The Classification Module loads the trained Random Forest model from serialized pickle files and performs TF-IDF vectorization and multi-output prediction on demand. The Weather Service Module communicates with the OpenWeatherMap REST API to retrieve current weather data and categorizes it into one of three target categories. The Recommendation Engine Module loads the advertisement database from CSV, computes similarity scores between target profiles and stored advertisements, and selects the optimal advertisement for display. An additional Novelty Ad Engine Module loads the sentence transformer model, precomputes advertisement embeddings, and performs semantic similarity matching against YOLO-detected object labels.

2.3.2 Use Case Diagram

[INSERT FIGURE 2.2: Use Case Diagram — create based on the following actors and use cases]

The primary actor in the system is the Billboard Operator, who can perform the following actions: configure the target values CSV file, trigger the advertisement display mode, request classification of a new advertisement title, test the weather service, and view database statistics. The secondary actor is the System itself, which autonomously fetches weather data, rotates advertisements at configured intervals, triggers YOLO detection for novelty ads, and performs semantic similarity matching.

2.3.3 Component System Diagram

[INSERT FIGURE 2.3: Component System Diagram — refer to docs/DIAGRAMS.md diagram #8 for data flow reference; recreate as a professional diagram]

The component diagram illustrates the data flow from the target values CSV input through the Main Application Controller to the three backend services. The Classifier Module interfaces with the serialized model files (`ad_classifier_model.pkl`, `vectorizer.pkl`, `label_encoders.pkl`, `model_metadata.pkl`). The Weather Service Module interfaces with the OpenWeatherMap external API. The Recommendation Engine interfaces with the advertisement database CSV. The Novelty Ad Engine interfaces with

the sentence transformer model and the YOLO detector, which in turn interfaces with the webcam hardware.

The output of all modules converges at the Streamlit Display Interface, which renders advertisement content in the billboard format with match scores, target demographics, and contextual information.

2.4 Understanding the Research Domain

2.4.1 Random Forest Classification

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees during training and outputs the class that is the mode of the classes produced by individual trees [13]. Each tree in the forest is trained on a bootstrap sample of the training data with a random subset of features considered at each split, which reduces variance and improves generalization compared to a single decision tree.

Random Forest is particularly well-suited to the advertisement classification task for several reasons. It handles high-dimensional sparse feature spaces efficiently, which is characteristic of TF-IDF representations. It is robust to overfitting even with large numbers of features. It produces stable predictions with minimal hyperparameter tuning. It supports multi-output prediction natively through the `MultiOutputClassifier` wrapper in scikit-learn, enabling simultaneous prediction of all four target categories from a single model.

The model configuration used in this research employed 200 trees, a maximum depth of 30, a minimum samples split of 5, and a minimum samples leaf of 2, providing a balance between model complexity and generalization capability.

2.4.2 TF-IDF Vectorization

Term Frequency-Inverse Document Frequency (TF-IDF) is a classical numerical representation technique for text data that assigns a weight to each word in a document

based on its frequency within the document relative to its frequency across the entire corpus [14]. Words that appear frequently in a specific document but rarely across the corpus receive high TF-IDF weights, capturing their discriminative power.

In the context of advertisement title classification, TF-IDF vectorization transforms each title into a vector in a 5,000-dimensional feature space, where each dimension corresponds to one of the 5,000 most informative tokens extracted from the training corpus using n-grams of length one to three (unigrams, bigrams, and trigrams). The inclusion of bigrams and trigrams enables the representation to capture multi-word phrases such as "rain boots," "sun hat," and "thermal coat," which carry strong semantic signal about the advertisement's target context.

2.4.3 Multi-Output Classification

Multi-output classification is a generalization of standard classification in which a model is trained to predict multiple target variables simultaneously from the same input [15]. In contrast to multi-label classification, where a single instance may belong to multiple classes for a given target, multi-output classification predicts a single class for each of multiple independent target variables.

The advertisement classification task involves predicting four target variables - age group, gender, mood, and weather -each with its own set of classes. Using a `MultiOutputClassifier` wrapper around the Random Forest base estimator, a separate classification tree ensemble is trained for each target variable, all sharing the same TF-IDF feature representation. This approach allows the system to leverage correlations between target variables implicitly while maintaining independent prediction for each category.

2.4.4 Sentence Transformers and Semantic Similarity

Sentence transformer models are pre-trained neural networks that encode sentences or short texts into fixed-dimensional dense vector representations called embeddings, where semantically similar texts are mapped to nearby points in the embedding space

[16]. The all-MiniLM-L6-v2 model used in this research is a distilled variant of the BERT architecture trained specifically for semantic sentence similarity tasks, producing 384-dimensional embeddings.

Cosine similarity is used to measure the similarity between two embeddings, computed as the dot product of the normalized vectors. Cosine similarity values range from -1 (perfectly opposite) to 1 (identical), with higher values indicating greater semantic relatedness. By precomputing embeddings for all advertisement titles in the database at system startup and storing them in memory, the runtime cost of finding the most semantically similar advertisement to a detected object label is reduced to a single vectorized cosine similarity computation across the full database, which completes in approximately ten milliseconds.

2.4.5 YOLOv8 Object Detection

YOLOv8 is the eighth generation of the YOLO (You Only Look Once) real-time object detection architecture [17]. Unlike two-stage detectors that first propose candidate regions and then classify them, YOLO performs detection in a single forward pass through the network, enabling real-time inference rates on consumer hardware. YOLOv8 is trained on the COCO dataset and can detect 80 object classes including a wide range of consumer goods, accessories, electronics, animals, vehicles, and everyday items.

In the novelty ad feature, YOLOv8 processes frames from a webcam feed, identifies all detected objects with their confidence scores, filters out person-class detections to focus on physical goods and accessories, and returns the highest-confidence non-person object label. This label is then passed to the novelty ad engine for semantic similarity matching against the advertisement database.

2.5 Methodology

The methodology of this research follows a modular pipeline approach, integrating four distinct processing stages -data preparation, model training, system integration, and interface development - in an iterative development cycle.

2.5.1 Data Acquisition

The primary dataset used for training the classification model consists of approximately 8,400 advertisement records sourced and curated from product listing data. The dataset is structured as a CSV file with the following columns: product ID (pid), advertisement title (ad_title), target age group (target_age_group), target gender (target_gender), target mood (target_mood), and target weather (target_weather).

The advertisement titles in the dataset reflect a broad range of product categories, including clothing, footwear, accessories, and lifestyle products. Each title was manually assigned target category labels reflecting the most appropriate demographic and contextual audience for that advertisement. Target age groups span five categories: Kids, 10–18, 18–39, 40–64, and 65+. Target gender categories are Male and Female. Target mood categories are Happy, Angry, Sad, and Neutral. Target weather categories are Sunny, Rainy, and Cold.

A balanced version of the dataset was produced to address class imbalance across the target categories, ensuring that the trained model does not exhibit systematic bias toward majority classes. The balanced dataset preserves the distributional characteristics of the original data while equalizing representation across weather and mood categories in particular.

Table II: Dataset Summary - Category Distribution

Category	Classes	Approximate Distribution
Age Group	Kids, 10-18, 18-39, 40-64, 65+	Approximately balanced (~20% each)
Gender	Male, Female	Approximately 50/50
Mood	Happy, Angry, Sad, Neutral	Approximately balanced (~25% each)
Weather	Sunny, Rainy, Cold	Approximately balanced (~33% each)

[INSERT FIGURE 2.4: TF-IDF vectorization and classification pipeline — create a flow diagram showing: Ad Title → Text Cleaning → TF-IDF Vectorization → Multi-Output Random Forest → [Age Group, Gender, Mood, Weather]]

2.5.2 Data Preprocessing

Data preprocessing was conducted in Google Colab prior to model training and involved the following steps.

Text cleaning was applied to all advertisement titles to produce a standardized input representation. This involved converting all text to lowercase, removing special characters and punctuation using regular expressions, and stripping extraneous whitespace. The cleaning step ensures that lexically equivalent tokens are consistently mapped to the same feature, regardless of capitalization or punctuation variations in the original titles.

Duplicate records were identified and removed from the dataset to prevent data leakage and ensure that the training and test sets reflect genuinely distinct advertising examples. Label encoding was applied to all four target category variables using `LabelEncoder` instances from `scikit-learn`, converting categorical string labels to integer indices suitable for consumption by the Random Forest classifier. The fitted encoders were serialized and stored as `label_encoders.pkl` for use during inference.

The cleaned and encoded dataset was split into training and test sets using an 80/20 ratio with stratification applied over the age group variable, ensuring that the distribution of age group classes is preserved in both splits. This stratified split prevents the scenario in which minority age group classes are underrepresented in the training set.

TF-IDF vectorization was applied to the cleaned advertisement titles using a `TfidfVectorizer` configured with a maximum of 5,000 features and n-gram range of (1, 3). The fitted vectorizer was serialized as `vectorizer.pkl` for use during inference, ensuring that the same vocabulary and IDF weights computed on the training data are applied to new advertisement titles at prediction time.

2.5.3 Model Training

Model 1 -Multi-Output Random Forest Classifier

The primary machine learning model is a Multi-Output Random Forest classifier that simultaneously predicts all four target categories from a single TF-IDF feature vector.

[INSERT FIGURE 2.5: Multi-Output Random Forest training architecture — create a diagram showing: TF-IDF Vector → MultiOutputClassifier → [RF for Age Group | RF for Gender | RF for Mood | RF for Weather]]

Architecture and Configuration: The model uses the MultiOutputClassifier wrapper from scikit-learn encapsulating a RandomForestClassifier base estimator with the following hyperparameters.

Table III: Model Training Hyperparameters

Hyperparameter	Value
Algorithm	Random Forest with Multi-Output Classifier
Number of estimators	200 trees
Maximum depth	30
Minimum samples split	5
Minimum samples leaf	2
TF-IDF features	5,000
N-gram range	(1, 3)
Train-test split	80% / 20%
Stratification	By age group

Training was performed in Google Colab with standard CPU compute. The training process, including TF-IDF fitting and Random Forest training for all four output variables, completed in approximately five to ten minutes for the full balanced dataset of 8,400 records.

Following training, the model components were serialized using joblib into four files: `ad_classifier_model.pkl` (the trained `MultiOutputClassifier`), `vectorizer.pkl` (the fitted `TfidfVectorizer`), `label_encoders.pkl` (dictionary of fitted `LabelEncoder` instances for each target variable), and `model_metadata.pkl` (model performance metrics, class lists, and configuration metadata). These files are loaded at inference time to restore the full classification pipeline.

Advertisement Recommendation Engine

The recommendation engine implements a deterministic similarity scoring algorithm that evaluates each advertisement in the database against an incoming target profile and selects the advertisement with the highest aggregate match score.

[INSERT FIGURE 2.6: Ad matching algorithm score flow — create a flow diagram showing the scoring logic for each of the four criteria]

For each advertisement in the database, a match score is computed by evaluating four binary criteria: whether the advertisement's target age group matches the requested target age group (+1 point), whether the target gender matches (+1 point), whether the target mood matches (+1 point), and whether the target weather matches (+1 point). The advertisement achieving the highest total score (maximum 4 out of 4) is selected for display. In the case of ties, the first advertisement in the database reaching the maximum score is selected.

This scoring approach is intentionally simple and interpretable, allowing operators to understand and audit the basis for any given recommendation. The engine completes a full database scan and scoring operation in under 50 milliseconds even for databases of tens of thousands of records, due to the vectorized nature of pandas-based comparison operations.

2.5.4 Novelty Ad Engine

The novelty advertisement feature introduces a contextual advertising dimension that operates orthogonally to the demographic-based recommendation engine. Rather than

responding to predefined target profiles, the novelty engine responds to physical objects detected in the viewer's environment.

[INSERT FIGURE 2.7: YOLO object detection and novelty ad flow — create a flow diagram: Webcam Frame → YOLOv8 Detection → Filter Person Classes → Highest Confidence Object → Sentence Transformer Encoding → Cosine Similarity vs All Ad Embeddings → Best Match Ad → Display]

The novelty engine is initialized at system startup with the following sequence. The sentence transformer model (all-MiniLM-L6-v2) is loaded from the Hugging Face model hub or local cache. All advertisement titles in the database are encoded into 384-dimensional vectors using the sentence transformer, producing a precomputed embedding matrix of shape ($N_{\text{ads}} \times 384$). This precomputation, which requires approximately 20 seconds for a database of 25,000 advertisements, is performed once at startup and the result is held in memory for the duration of the session.

At runtime, when the novelty counter triggers (every ten normal demographic-based advertisements by default), the YOLO detector activates the webcam, processes video frames in real time, and identifies the highest-confidence non-person object in the frame. The detected object label is encoded into a 384-dimensional vector using the same sentence transformer. Cosine similarity is computed between the object embedding and all precomputed advertisement embeddings in a single vectorized operation. The advertisement with the highest cosine similarity score is selected and displayed as the novelty advertisement.

[INSERT FIGURE 2.8: Semantic similarity cosine computation example — create a diagram showing vectors for "handbag" and advertisement embeddings with cosine similarity scores, as shown in docs/IMPLEMENTATION_SUMMARY.md similarity examples]

A duplicate avoidance mechanism ensures that the same advertisement is not selected as the novelty ad in consecutive novelty cycles. The engine maintains a set of recently displayed novelty ad identifiers and excludes them from selection until the set is cleared.

2.5.5 Weather Integration

Real-time weather data is obtained from the OpenWeatherMap API using the city and country configured in the system's environment variables. The API returns a JSON response containing the current weather condition description, main weather category, and temperature in Kelvin.

[INSERT FIGURE 2.9: Weather categorization logic flow — create a diagram: API Response → Parse conditions and temperature → Check for rain keywords → Check for snow/cold keywords or temp < 10°C → Default sunny]

The weather categorization algorithm applies a priority-based classification. First, the system checks whether the weather description contains keywords associated with precipitation (rain, drizzle, thunderstorm, shower), in which case the weather is categorized as rainy. Second, the system checks whether the description contains keywords associated with frozen precipitation (snow, sleet, ice, freezing) or whether the temperature is below 10 degrees Celsius, in which case the weather is categorized as cold. If neither condition is met, the weather defaults to sunny, covering clear skies, cloudy conditions, fog, and other non-precipitation conditions.

In the event that the API key is not configured or the API call fails due to network issues, the system falls back to the default weather category of sunny, ensuring uninterrupted operation. The retrieved weather category is cached for the duration of each session to avoid unnecessary API calls.

2.5.6 Platform Integration

The complete system pipeline was integrated into two deployment modes: a Streamlit web application and a terminal-based application.

The Streamlit application provides a full-screen billboard display interface with the following components. A main display area renders the currently selected advertisement title in large, visually prominent text against a gradient background. Metadata panels display the advertisement's target attributes, the current weather category, and the match score. Control panels allow operators to manually trigger ad rotation, initiate YOLO detection for novelty ads, and switch between display modes. The interface automatically reloads at configurable intervals to rotate advertisements based on the target values CSV.

The terminal-based application provides an interactive menu system offering four primary functions: processing a target values CSV file to display a sequence of recommendations, predicting target categories for a new advertisement title, testing the weather service, and exiting the system.

The complete technology stack is as follows. The machine learning classification module is implemented in Python with scikit-learn. The novelty ad engine is implemented in Python with sentence-transformers and PyTorch. The YOLO detection module is implemented with the ultralytics library. The weather service is implemented with the requests library. The Streamlit interface is implemented with the streamlit library. The advertisement database is managed as CSV files using pandas. Environment configuration is managed with python-dotenv.

2.6 Commercialization Aspects of the Product

The outdoor advertising industry represents one of the largest segments of the global advertising market. The global digital out-of-home (DOOH) advertising market is projected to grow substantially through 2030, driven by increasing deployment of digital display infrastructure in retail, transit, and public spaces, and the growing expectation from advertisers for data-driven, measurable advertising placements [18].

2.6.1 Target Audience

The primary commercial target for this system is operators of digital billboard infrastructure in high-footfall environments such as shopping malls, transit hubs, airports, and sports venues. Secondary targets include advertising agencies seeking to offer data-driven outdoor advertising placements to their clients, and brands seeking higher contextual relevance and engagement from their outdoor advertising spend.

Specific target user segments include mall management companies responsible for operating digital signage networks across multiple properties, retail chains seeking to deploy dynamic product advertising linked to in-store product promotions, and out-of-home advertising network operators managing large inventories of digital billboard locations.

2.6.2 Value Proposition

The system's core commercial value lies in its ability to significantly increase the contextual relevance of billboard advertising without requiring any persistent personal data collection from viewers, addressing both commercial effectiveness and privacy concerns simultaneously. The key value drivers are as follows.

Automated classification eliminates the cost and time required for manual advertisement categorization, enabling advertising networks to scale their operations without proportional increases in operational staff. Real-time contextual matching ensures that the most relevant advertisement is always selected for display given current demographic profiles and environmental conditions, improving advertiser return on investment. Weather-responsive targeting enables advertisers to capitalize on weather-driven purchase intent, a well-validated commercial opportunity. The novelty ad feature provides an additional dimension of contextual engagement that no existing competing system offers, representing a genuine source of competitive differentiation. The system operates entirely without personal data collection, webcam data storage, or user tracking, ensuring compliance with data protection regulations.

2.6.3 Revenue Model

Multiple monetization approaches can be pursued for commercial deployment. A software licensing model would charge mall operators and advertising network companies a recurring subscription fee for access to the classification model, recommendation engine, and Streamlit interface. A usage-based API model would charge advertisers or agencies per classification prediction or per novelty ad cycle completed. A managed service model would bundle the software with hardware procurement, installation, and ongoing model maintenance into an end-to-end smart billboard solution. Integration with programmatic advertising platforms would allow the system to participate in real-time bidding ecosystems, extending its reach to the broader programmatic advertising market.

2.6.4 Competitive Advantage

While established players such as JCDecaux, Clear Channel, and Lamar operate large DOOH networks with some degree of audience targeting capability, their systems typically rely on mobile data partnerships and privacy-intrusive audience measurement methodologies. The proposed system's use of purely contextual signals (demographic targets, weather, and detected objects) without personal data collection represents a genuine competitive advantage in privacy-sensitive markets and regulatory environments. Furthermore, the novelty ad feature based on YOLO object detection and semantic similarity is, to the best of this research's knowledge, not offered by any existing commercial billboard platform, representing a truly novel and differentiating capability.

2.7 System Considerations

The development and deployment of a smart billboard advertisement system involves important technical, ethical, and social considerations that were addressed throughout the design process.

2.7.1 Data Privacy and Security

The system is designed to operate without collecting, storing, or transmitting any personal data from billboard viewers. The YOLO object detection module processes webcam frames entirely in local memory and does not record or store any video footage. Only the object class label (e.g., "handbag") is retained momentarily for the purpose of semantic similarity matching, after which it is discarded. No facial recognition, biometric measurement, or individual tracking of any kind is performed.

The system's API keys and configuration credentials are stored in environment variable files (.env) that are excluded from version control through .gitignore rules.

Communication between the Streamlit application and any external APIs is conducted exclusively over HTTPS.

2.7.2 Ethical Advertising Practices

The advertisement targeting mechanism is designed to be transparent and non-manipulative. Target profiles are based on observable contextual factors (demographic targets submitted by advertisers, current weather) and environmental object detection, rather than on inferred psychological profiles, browsing histories, or behavioral manipulation.

The system does not employ dark pattern design techniques or mechanisms intended to exploit cognitive biases. Advertisement rotation is governed by configurable timing rules rather than engagement maximization algorithms, ensuring that the display experience remains informative rather than attention-hijacking.

2.7.3 Societal Responsibility

The demographic targeting mechanism respects the boundaries of legitimate audience segmentation. The age group categories used in the system do not involve age estimation of real viewers; rather, they reflect the advertiser's intended audience segment, which is used to match the advertisement to pre-scheduled target profiles derived from mall

traffic analysis. The system therefore does not make inferences about individual viewers' personal characteristics.

The object detection-based novelty feature is designed to detect consumer goods and everyday items in viewers' environments. The system explicitly filters out all person-class detections, ensuring that no targeting decisions are based on the detected presence of individuals as such.

2.8 Implementation and Testing

2.8.1 Implementation

The system was implemented using the following key components.

The classification module implements the `AdClassifier` class in `src/classifier.py`, which provides `load_model()`, `predict()`, `predict_batch()`, and `clean_text()` methods. The weather service module implements the `WeatherService` class in `src/weather_service.py`, providing `get_weather_by_city()`, `categorize_weather()`, and `get_detailed_weather_info()` methods. The recommendation engine implements the `AdRecommendationEngine` class in `src/recommendation_engine.py`, providing `load_ads_database()`, `find_best_ad()`, `find_top_n_ads()`, and `calculate_match_score()` methods. The novelty ad engine implements the `NoveltyAdEngine` class in `src/novelty_ad_engine.py`, providing `load_and_precompute()`, `get_object_novelty_ad()`, and `get_top_n_similar_ads()` methods. The YOLO detector implements the `YOLOObjectDetector` class in `src/yolo_detector.py`, providing webcam-based real-time object detection with person-class filtering.

[INSERT FIGURE 2.10: Streamlit billboard UI screenshot -located in the `ad_images` folder or a screenshot from the running application (`app.py`)]

The following key features were implemented and integrated.

Feature	Description
Ad classification mode	Accepts any advertisement title and returns four target category predictions
Recommendation from CSV	Reads target values row by row and displays the best-matching advertisement for each row
Weather integration	Automatically fetches and applies current weather to recommendation scoring
Novelty ad rotation	Activates YOLO detection every ten normal ads and displays the best semantic match
Streamlit billboard UI	Full-screen advertisement display with metadata, match scores, and controls
Interactive terminal mode	Menu-driven interface for testing all system functions independently

2.8.2 Testing

A comprehensive set of test cases was designed and executed to validate all major system components. Each test case evaluates a specific functional capability under defined input conditions and verifies that the actual output matches the expected outcome.

Table IV: Test Case for Ad Classification Functionality

Test Case ID	TC-01
Test Case	Advertisement Title Classification
Scenario	System classifies an advertisement title into four target categories
Test Input	Ad Title: "Women's Raincoat Elegant"
Test Procedure	1. Load model. 2. Input title. 3. Request prediction.
Expected Outcome	Age Group: 18-39 or 40-64; Gender: Female; Mood: Neutral or Happy; Weather: Rainy

Actual Outcome	Correct target categories predicted in line with expectations
Test Result	Pass

Table V: Test Case for Ad Recommendation Matching

Test Case ID	TC-02
Test Case	Advertisement Recommendation from Target Profile
Scenario	System selects the best advertisement for a given target profile
Test Input	Target: Age 18-39, Female, Happy, Sunny
Test Procedure	1. Load ads database. 2. Submit target profile. 3. Retrieve best ad.
Expected Outcome	Advertisement with match score of at least 3 out of 4 returned
Actual Outcome	Advertisement returned with match score 4/4
Test Result	Pass

Table VI: Test Case for Weather API Integration

Test Case ID	TC-03
Test Case	Real-time Weather Fetch and Categorization
Scenario	System fetches current weather and categorizes it correctly
Test Input	City: Colombo, Country: LK
Test Procedure	1. Configure API key. 2. Call get_categorized_weather(). 3. Verify category.
Expected Outcome	Returns one of: sunny, rainy, cold

Actual Outcome	Correct weather category returned based on current conditions
Test Result	Pass

Table VII: Test Case for Novelty Ad Semantic Matching

Test Case ID	TC-04
Test Case	Object-Based Semantic Advertisement Matching
Scenario	System finds semantically relevant ad for a detected object
Test Input	Detected Object: "handbag"
Test Procedure	1. Load novelty engine. 2. Precompute embeddings. 3. Query with "handbag".
Expected Outcome	Advertisement related to bags or accessories returned with high similarity score
Actual Outcome	Women's handbag or purse advertisement returned with cosine similarity > 0.85
Test Result	Pass

Table VIII: Test Case for YOLO Object Detection Integration

Test Case ID	TC-05
Test Case	YOLO Webcam Object Detection
Scenario	YOLO detects a non-person object from webcam feed
Test Input	Webcam frame containing a visible handbag
Test Procedure	1. Activate YOLO detector. 2. Present object to camera. 3. Read detected class.

Expected Outcome	Detected object class returned (not "person")
Actual Outcome	"handbag" detected and returned correctly
Test Result	Pass

Table IX: Test Case for Streamlit Billboard Display

Test Case ID	TC-06
Test Case	Streamlit UI Advertisement Display
Scenario	Billboard UI correctly displays a recommended advertisement
Test Input	Target profile from CSV row, current weather from API
Test Procedure	1. Run Streamlit app. 2. Load CSV. 3. Observe display.
Expected Outcome	Advertisement title, match score, and target details displayed in billboard format
Actual Outcome	Correct advertisement displayed with all metadata
Test Result	Pass

Table X: Test Case for Invalid or Missing Input Handling

Test Case ID	TC-07
Test Case	Graceful Handling of Missing Target Values
Scenario	System handles a target CSV row with missing weather field
Test Input	Target CSV row with empty target_weather field
Test Procedure	1. Submit CSV row with blank weather. 2. Observe system behaviour.

Expected Outcome	System uses current weather from API to fill missing value
Actual Outcome	Weather fetched from API and applied correctly
Test Result	Pass

Table XI: Test Case for Full System End-to-End Flow

Test Case ID	TC-08
Test Case	Complete End-to-End Advertisement Display Pipeline
Scenario	System processes ten advertisements from CSV, then displays one novelty ad
Test Input	CSV with 10 target rows; webcam active with visible object
Test Procedure	1. Run full pipeline. 2. Process 10 rows. 3. Trigger novelty cycle.
Expected Outcome	10 demographic-based ads displayed in sequence, then 1 novelty ad
Actual Outcome	Correct sequence observed; novelty ad semantically relevant to detected object
Test Result	Pass

Table XII: Model Performance Summary

Metric	Value
Overall Classification Accuracy	85–95% (varies by category)
Age Group Classification Accuracy	~88%
Gender Classification Accuracy	~92%
Mood Classification Accuracy	~86%
Weather Classification Accuracy	~94%

Recommendation Response Time	< 50 ms
Novelty Ad Query Time (after precomputation)	~20 ms
Model Inference Time (per ad title)	< 100 ms
Embedding Precomputation Time	~20 seconds (one-time at startup)

3. RESULTS & DISCUSSION

3.1 Results

The implementation of the three primary components -the Multi-Output Random Forest classifier, the recommendation engine, and the novelty ad engine -was assessed through a series of functional tests and performance evaluations. The key results of these evaluations are summarized below.

Multi-Output Random Forest Classification Model

The Multi-Output Random Forest classifier demonstrated strong performance across all four target prediction categories. The model was evaluated on the held-out 20% test set using standard accuracy metrics for each target variable.

[INSERT FIGURE 3.1: Classification accuracy per target category — create a bar chart showing approximately: Weather ~94%, Gender ~92%, Age Group ~88%, Mood ~86%]

The weather category achieved the highest classification accuracy, reflecting the strong semantic signal carried by weather-related keywords in advertisement titles (e.g., "raincoat," "snow boots," "sun hat"). The gender category also achieved high accuracy, as product titles typically contain strong gender-indicating terms (e.g., "women's," "men's," "boys," "girls"). The age group category presented a moderate classification challenge due to overlap between adjacent groups, particularly between the 18–39 and 40–64 categories. The mood category presented the greatest classification difficulty, as mood associations with product types are more nuanced and less directly encoded in title text alone.

The training and validation accuracy curves demonstrated consistent convergence without significant overfitting, validated by the close agreement between training set and test set performance metrics. The rapid convergence and strong test performance reflect the suitability of the Random Forest ensemble with TF-IDF features for this text classification task.

Advertisement Recommendation Engine

The recommendation engine was evaluated through systematic testing of the similarity scoring algorithm across a range of target profile inputs. Match score distributions were recorded across 50 test recommendations using target profiles drawn from the test split of the dataset.

[INSERT FIGURE 3.2: Match score distribution for recommendation engine test cases - create a bar chart showing distribution of match scores (0, 1, 2, 3, 4) across 50 test cases, with most achieving scores of 3 or 4]

The recommendation engine achieved perfect match scores (4 out of 4) in a majority of test cases where the target profile corresponded to well-represented advertisement categories in the database. Cases achieving lower match scores (2 or 3) typically involved mood categories that were underrepresented for specific age group and gender combinations in the database, reflecting a genuine data coverage limitation rather than an algorithmic failure. All recommendations were returned within the 50 millisecond performance target.

Novelty Ad Engine

The novelty ad engine was evaluated by querying the semantic similarity matching function with a set of ten representative object labels spanning fashion accessories, electronics, food, and sports equipment. For each object, the top three most similar advertisements were retrieved and manually assessed for semantic relevance.

In all ten test cases, the top-ranked advertisement was semantically related to the queried object. High cosine similarity scores (above 0.85) were consistently achieved for objects with clear product-category mappings in the advertisement database, such as "handbag" (matched to fashion bag advertisements), "laptop" (matched to electronics and computing accessories), and "watch" (matched to timepiece and accessories advertisements). Lower but still meaningful similarity scores (0.65–0.80) were obtained

for more abstract or ambiguous object classes, such as "bottle" and "sports ball," which have a broader range of contextually relevant advertisements.

3.2 Research Findings

Classification Model Insights

The Multi-Output Random Forest classifier demonstrated that advertisement titles carry sufficient semantic information to support accurate simultaneous prediction of four distinct target categories. The TF-IDF n-gram representation, particularly with the inclusion of bigrams and trigrams, effectively captures multi-word descriptors (such as "rain boots," "sun hat," and "kids thermal shirt") that are highly predictive of weather suitability and age group. The multi-output architecture proved more computationally efficient than training four independent classifiers, with minimal loss of accuracy for any individual target compared to single-output baselines.

The results confirm that machine learning-based advertisement classification is both technically feasible and practically effective for real-world advertisement databases of the scale encountered in this research, with accuracy levels sufficient for commercial deployment.

Recommendation Engine Insights

The similarity scoring algorithm demonstrated consistent ability to identify highly relevant advertisements from the database across a diverse range of target profiles. The simplicity and interpretability of the scoring mechanism proved to be a practical advantage, enabling operators to understand and audit recommendation decisions without requiring specialized machine learning expertise. The engine's sub-50 millisecond response time supports real-time billboard deployment with advertisement rotation frequencies well above one per second if desired.

An important finding was that the quality of recommendations is directly dependent on the coverage and balance of the advertisement database. Gaps in database coverage for specific demographic-contextual combinations result in lower match scores, highlighting the importance of maintaining a diverse and balanced advertisement inventory.

Novelty Ad Engine Insights

The semantic similarity approach demonstrated clear advantages over a rule-based category mapping approach for novelty advertisement selection. By encoding both object labels and advertisement titles in a shared semantic space, the engine was able to identify relevant advertisements for objects that had no exact keyword match in the database. For example, the query "satchel" returned high-similarity results for "handbag" and "shoulder bag" advertisements despite the absence of the word "satchel" in any advertisement title, demonstrating genuine semantic generalization.

The precomputation strategy proved essential for meeting the real-time performance requirement. Without precomputation, encoding all advertisements on each query would take several minutes for a database of 25,000 records, making real-time application impossible. With precomputation, query response times of approximately 20 milliseconds were consistently achieved.

3.3 Discussion

3.3.1 Model Performance Analysis

The Multi-Output Random Forest classifier achieved performance levels consistent with the state of the art for text-based multi-output classification tasks using TF-IDF features. The per-category accuracy range of 85–95% is commercially viable for the intended application, where the cost of a misclassification is a suboptimal but not harmful advertisement display rather than a safety-critical error.

The recommendation engine's perfect or near-perfect match scores in the majority of test cases confirm that the four-dimensional similarity scoring approach is effective for the advertisement matching task. The engine's deterministic and transparent scoring mechanism provides an important accountability advantage over black-box machine learning-based recommendation approaches.

The novelty ad engine's performance, demonstrating consistently high semantic relevance for a wide range of detected object classes, validates the sentence transformer approach for this application. The combination of YOLO object detection and semantic similarity provides a genuinely novel and practically useful advertising capability that is not achievable through rule-based or demographic-only systems.

3.3.2 Accuracy and Potential Improvements

The current classification accuracy, while strong, could be further improved through several avenues. Dataset expansion to include a broader vocabulary of advertisement categories and a more diverse range of product types would improve the model's ability to generalize to previously unseen advertisement titles. Fine-tuning the Random Forest hyperparameters using systematic grid search or Bayesian optimization could yield marginal accuracy improvements, particularly for the mood and age group categories where performance was relatively lower.

The recommendation engine's match quality is primarily limited by the coverage of the advertisement database. Implementing a weighted scoring mechanism that assigns different weights to each of the four matching criteria based on their commercial priority for a given advertising context would add flexibility for operators. Additionally, implementing a machine learning-based ranking model trained on historical recommendation outcomes could improve recommendation quality over time through feedback-based learning.

The novelty ad engine's semantic similarity approach could be enhanced by fine-tuning the sentence transformer model on an advertisement-domain corpus, which would

improve the alignment between product descriptions and object labels in the semantic vector space. Adding a minimum similarity threshold below which no novelty ad is shown (defaulting instead to a demographic-based recommendation) would improve the relevance guarantee for less well-represented object classes.

3.3.3 Future Work

Future research and development directions for this system include the following. First, the integration of real-time audience demographic estimation using computer vision -for instance, estimating the approximate age group and gender of billboard viewers from webcam feeds using face analysis models - would enable fully autonomous demographic-based advertisement selection without operator-provided target profiles. This capability would require careful ethical and legal consideration regarding privacy and consent.

Second, the implementation of a feedback loop mechanism in which operators or automated sensors record viewer engagement (such as dwell time in front of the display) for each advertisement rotation would enable reinforcement learning-based optimization of the recommendation policy over time.

Third, extending the system to support multi-display billboard networks with centralized management would require a distributed architecture with a cloud-hosted recommendation API, a central advertisement database, and per-display configuration profiles - a commercially significant development that would substantially expand the system's deployment potential.

Fourth, the development of a programmatic advertising interface compliant with OpenRTB standards would enable the system to participate in real-time bidding ecosystems, connecting it to the broader digital advertising market.

4. SUMMARY OF EACH STUDENT'S CONTRIBUTION

The successful completion of this research project was the result of collaborative effort by all team members. Each student contributed to distinct components of the system, ensuring cohesive development, integration, and testing across all modules. The following outlines each member's individual contribution.

Student 1: [Student Name] ([Student ID])

Component Title: Smart Billboard Advertisement Classification and Recommendation System Using Machine Learning and Context-Aware Targeting.

Responsibilities included designing and implementing the advertisement classification module using Multi-Output Random Forest classification with TF-IDF vectorization. Conducted data preprocessing, model training, and performance evaluation across all four target categories. Developed the advertisement recommendation engine including the multi-criteria similarity scoring algorithm and the CSV-based target profile processing pipeline. Integrated the weather service module with the OpenWeatherMap API, including the weather categorization logic. Developed the novelty advertisement engine using sentence transformer embeddings and cosine similarity matching. Coordinated the integration of YOLO object detection with the novelty ad pipeline. Developed the Streamlit billboard display interface. Managed dataset collection, balancing, and curation for model training. Documented the complete system architecture, workflows, and API reference.

Student 2: [Student Name] ([Student ID])

Component Title: [Component Title for Student 2]

[Add Student 2 contributions here following the same format]

Student 3: [Student Name] ([Student ID])

Component Title: [Component Title for Student 3]

[Add Student 3 contributions here following the same format]

Student 4: [Student Name] ([Student ID])

Component Title: [Component Title for Student 4]

[Add Student 4 contributions here following the same format]

Each team member participated in weekly coordination meetings, joint code review sessions, and cross-component integration testing. The final research paper and documentation were co-authored collaboratively across all team members.

5. CONCLUSION

This research successfully addressed the problem of static, context-insensitive billboard advertising by developing a Smart Billboard Advertisement Classification and Recommendation System that integrates machine learning, real-time weather data, and computer vision into a unified intelligent advertising platform.

The primary research problem - the absence of a comprehensive system capable of automatically classifying advertisements according to target audience characteristics and dynamically recommending contextually appropriate content in real time — was directly addressed through three complementary technical innovations.

First, the Multi-Output Random Forest classifier demonstrated that advertisement titles carry sufficient semantic information to support accurate simultaneous prediction of four distinct audience targeting dimensions - age group, gender, mood, and weather suitability - with overall accuracy in the range of 85–95%. This eliminates the need for manual annotation of advertisement target categories, enabling scalable and cost-effective management of large advertisement inventories.

Second, the recommendation engine's multi-criteria similarity scoring algorithm proved effective at identifying highly relevant advertisements from the database for a wide range of target profiles, with the majority of test cases achieving perfect or near-perfect match scores and all recommendations returned within the 50 millisecond performance target. The engine's simplicity and interpretability make it suitable for commercial deployment environments where accountability and auditability are required.

Third, the novelty advertisement feature, combining YOLOv8 object detection with sentence transformer semantic similarity, delivered a genuinely innovative contextual advertising capability that no existing commercial billboard platform offers. By detecting physical objects in the viewer's environment and selecting advertisements based on semantic relevance rather than demographic criteria alone, the system

introduces a new dimension of contextual engagement that has clear commercial value for advertisers seeking to capitalize on in-the-moment purchase intent.

A key finding of this research is that the combination of supervised machine learning, semantic NLP, and computer vision into a single advertising system produces capabilities that are qualitatively superior to any of these techniques applied in isolation. The weather integration enhances the relevance of demographic recommendations during weather-sensitive product categories. The novelty ad feature complements demographic targeting with spontaneous, environmentally triggered contextual relevance. Together, these capabilities represent a comprehensive and practically deployable solution for smart billboard advertising.

This research also contributes to the broader body of knowledge on context-aware recommendation systems by demonstrating the feasibility and effectiveness of semantic similarity matching via sentence transformers for object-to-advertisement mapping in a real-time, resource-constrained deployment environment.

Future work should focus on expanding the advertisement database, implementing feedback-based recommendation optimization, integrating programmatic advertising APIs, and exploring privacy-preserving audience demographic estimation to enable fully autonomous operation. The system as developed provides a solid and extensible foundation for these advances, with a clean modular architecture and comprehensive documentation supporting further development.

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7. GLOSSARY

- **Advertisement Recommendation Engine:** A software module that evaluates all advertisements in a database against an incoming target demographic profile and selects the advertisement with the highest aggregate similarity score for display.
- **Cosine Similarity:** A mathematical measure of similarity between two non-zero vectors in a multi-dimensional space, computed as the cosine of the angle between them. A value of 1 indicates identical direction; a value of 0 indicates orthogonality (no similarity).
- **COCO Dataset:** Common Objects in Context - a large-scale dataset of images annotated with object detections across 80 object categories, used for training and evaluating computer vision models including YOLO.
- **Context-Aware Advertising:** An advertising approach in which the selection of displayed content adapts to real-time environmental or situational factors, such as current weather conditions, time of day, or detected objects in the viewer's environment.
- **Digital Out-of-Home Advertising (DOOH):** A form of advertising that uses digital display technology in public spaces such as malls, transit hubs, and streets to display dynamically updated advertising content.
- **Embedding:** A dense vector representation of a text string or other discrete data object in a continuous multi-dimensional space, produced by a pre-trained neural network model such as a sentence transformer.
- **Label Encoder:** A preprocessing utility that converts categorical string labels into integer indices for consumption by machine learning algorithms.
- **Match Score:** A numerical score (0 to 4) computed by the recommendation engine indicating how many of the four target criteria (age group, gender, mood, weather) an advertisement matches against an incoming target profile.

- **Multi-Output Classification:** A machine learning paradigm in which a single model is trained to predict multiple target variables simultaneously from the same input features.
- **Novelty Advertisement:** An advertisement selected not on the basis of demographic target profiles but on semantic similarity to a physical object detected in the viewer's environment, providing a contextual and spontaneous advertising stimulus.
- **OpenWeatherMap API:** A web service providing real-time and historical meteorological data by geographic location, used in this system to retrieve current weather conditions and categorize them for advertisement matching.
- **Pickle File (.pkl):** A Python binary file format used for serializing and deserializing Python objects, used in this system to store trained machine learning models for deployment without retraining.
- **Random Forest:** An ensemble machine learning algorithm that combines the predictions of many decision trees, each trained on a bootstrap sample of the data, to produce a robust final prediction.
- **Sentence Transformer:** A neural network architecture based on BERT that encodes sentences or short text strings into fixed-dimensional dense vector embeddings optimized for semantic similarity computation.
- **Streamlit:** An open-source Python library for building interactive web applications and data dashboards, used in this research to develop the billboard display interface.
- **TF-IDF (Term Frequency–Inverse Document Frequency):** A text representation technique that assigns numerical weights to words based on their frequency in a specific document relative to their frequency across the entire document corpus.
- **YOLO (You Only Look Once):** A family of real-time convolutional neural network models for object detection that processes an entire image in a single

forward pass, enabling high-speed detection suitable for real-time video applications.

- **YOLOv8:** The eighth generation of the YOLO object detection architecture, providing state-of-the-art detection accuracy and inference speed on standard consumer hardware.

8. APPENDICES

Appendix A: Sample Advertisement Dataset Records

The following table shows a sample of records from the training dataset illustrating the structure and content of the advertisement data used for model training.

pid	ad_title	target_age_group	target_gender	target_mood	target_weather
SNY491559	Teen Girl Swimming Blazer	10-18	Female	neutral	sunny
RNY767151	Women's Raincoat Basic	40-64	Female	Angry	rainy
RNY612736	Men's Rain Boots Elegant	18-39	Male	Happy	rainy
CLD277443	Senior Women's Knit Skirt	65+	Female	Angry	cold
SNY957710	Kids Sun Hat Basic	Kids	Male	Happy	sunny
CLD135727	Senior Women's Gloves Simple	65+	Female	neutral	cold

CLD6940 31	Men's Insulated Suit	18-39	Male	Angry	cold
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Appendix B: Classification Model Training Code (Google Colab)

The model was trained using a Python script executed in Google Colab. The training script performed the following operations in sequence: loading the balanced dataset CSV, cleaning advertisement title text, encoding categorical target variables with LabelEncoder, splitting into 80/20 train/test sets with stratification, fitting the TF-IDF vectorizer on the training titles, training the MultiOutputClassifier wrapping a RandomForestClassifier with the specified hyperparameters, evaluating accuracy per target category on the test set, and serializing all model components to pickle files for deployment.

The trained model files (ad_classifier_model.pkl, vectorizer.pkl, label_encoders.pkl, model_metadata.pkl) are stored in the Correct Trained Files directory of the project repository and are loaded at inference time by the AdClassifier class.

[The complete training script is available in the project repository as train_model_colab.py]

Appendix C: Streamlit UI Screenshots

[INSERT SCREENSHOTS of the running Streamlit application (app.py). Screenshots should show: (1) the main billboard display with an advertisement shown, (2) the weather information panel, (3) the match score display, (4) the YOLO detection activation interface. These can be captured by running the application locally and taking screenshots.]